

The Association between Concentrations of Green Tea and Blood Glucose Levels

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Our objective was to examine whether habitual green tea consumption is associated with blood glucose levels and other biomarkers of glucose metabolism. We conducted a cross-sectional study of 35 male volunteers, 23–63 years old and residing in Shizuoka Prefecture in Japan. Biochemical data were measured and we conducted a questionnaire survey on health, lifestyle, and nutrition, as well as frequency of consumption and concentrations (1%, 2%, and 3%) of green tea. Men who consumed a 3% concentration of green tea showed lower mean values of fasting blood glucose and fructosamine than those who consumed a 1% concentration. Fasting blood glucose levels were found to be significantly associated with green tea concentration ($\beta = -0.14$, $p = 0.03$). However, green tea consumption frequency showed no significant differences in mean levels of blood glucose, fructosamine and hemoglobin A1c. In conclusion, our findings suggest that the consumption of green tea at a high concentration has the potential to reduce blood glucose levels.